

## Math 125 Midterm Review 2

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Solutions on the following page.

1. Fall 2015 Q4
2. Fall 2015 Q5
3. Fall 2018 Q2 (parts a-b)
4. Fall 2018 Q5

1. Fall 2015 Q4

(a)  $y'(x) = (6x - y^2 \sin(xy^2))/(2xy \sin(xy^2) - 1)$

(b)  $y = 6 - 6x$  or  $y - 0 = -6(x - 1)$

2. Fall 2015 Q5

(a) Show  $f$  has at least one zero by using IVT.

(b) Show  $f$  has at most one zero by computing  $f'(x) = 3x^2 + 2 - \cos x$  and noting this is positive, so  $f$  is increasing everywhere and cannot have more than one zero.

3. Fall 2018 Q2 (parts a-b)

(a) Using quotient rule,  $f'(x) = \frac{x \sec^2 x - \tan x}{x^2}$

(b) Using product rule,  $f'(x) = 8x(x^2 + 1)^3 - 3((x^2 + 1)^4)(1 - x)^2$

4. Fall 2018 Q5

(a)  $\frac{dy}{dx} = \frac{1 - y \cos xy}{x \cos xy + \sin y}$

(b)  $y = -(1 + \pi)x + 2\pi + 1$  or  $y - \pi = -(1 + \pi)(x - 1)$